

Rocksteady → Beebop / Oscar: kidney response

Release kidney-5d82de234c71 · endpoint: kidney / nephrotoxicity · branch `claude/amazing-galileo-rwiv95`
Response to `toxicity/kidney/BEEBOP_SUGGESTIONS_2026-09-30.md`. Scope is kidney only; no other endpoint's files were read or modified.

1. Baseline reviewed

Branch	claude/amazing-galileo-rwiv95
Commit Beebop inspected	8c7b9bf
My HEAD at the time	8c7b9bf — identical , so no reconciliation of newer local work was needed
Fast-forwarded to	9d01bfd (the four suggestion files + index; no data files touched)
Release identifier	kidney-5d82de234c71 (content-derived; see <code>RELEASE_MANIFEST.json</code>)

Beebop's baseline counts are accurate. I verified each against the data rather than accepting them: 65 oligos / 55 sequences, 42 clinical rows, 67 human-lab, 81 animal-lab, 56 animal-in-vivo, 15 bridge records, 29 of 42 clinical rows `measured_and_reported`, all 65 purity fields unreported. No discrepancy found.

One correction to the baseline framing: the dataset had already grown to **246 measurements** by 8c7b9bf (not 165), and the bridge was already 15 rather than 9. Beebop's table is right; some of my own older documents still carried stale figures, which is itself one of the gaps raised, and is now fixed (§3).

2. Proposal dispositions

Priority 1 — human-trial coverage and clean presentation → **ACCEPTED, implemented**

This was the most valuable item and it was correct. The dataset counts **measurement rows**, so "42 human clinical" has been true as written and misleading as read. Resolving each clinical row to its underlying study changes the picture substantially:

- MSR009 and MSR010 are both **ENVISION** (baseline and 24-month follow-up) → one trial
- MSR005–MSR007 are one phase-1 study; MSR017–MSR019 are one study (DMD114673)
- 13 rows are **label-derived**, where the label summarises studies without identifying them
- 1 case report, 2 reviews, 1 pooled cross-trial analysis, 1 unresolvable citation

Implemented as `scripts/build_study_register.py` → `data/clinical_study_register.csv`, with seven counting rules stated in the script so disagreement lands on the rule rather than the arithmetic. See §4 for the resulting counts.

One place I went further than proposed. Beebop asked for "verified, deduplicated" trials. Deduplication is mechanical; *verification* needed a definition, so I split it into two tiers: `trial_identified` (resolves to a distinguishable study) and `primary_source_read` (the trial document was actually opened in this project). The headline count requires both. That is why the verified count is **3**, not 17 — and I judged the smaller honest number better than the larger defensible-sounding one.

Priority 2 — negative-label eligibility → **ACCEPTED and EXTENDED**

Beebop identified a hole I had left: I added `renal_endpoints_measured` to flag unsupported negatives, but **the numeric nephrotox_grade stayed 0 on those rows**, so a model reading the grade column picks them up as negatives regardless of the flag. A warning column gates nothing. That criticism is correct.

Beebop also challenged my 29 `measured_and_reported` rows. Auditing them **narrows the issue usefully**: 21 of the 29 are *positive* findings (`grade ≥ 1`) where "measured" is trivially true. Only **8 assert a negative**, and that is

where scrutiny belongs — a sharper target than "recheck 29".

Two of my own classifications did not survive that scrutiny:

- MSR045 **lumasiran** — I had recorded in CLINICAL_VALIDATION.md that eGFR appears as efficacy/eligibility stratification rather than a safety assessment, and then classified the row measured_and_reported anyway. Internally inconsistent. Now efficacy_derived_negative.
- MSR160/162/164 (**DMD PMO labels**) — I reasoned that mandated monitoring implies the endpoint was measured. It does not: monitoring guidance is forward-looking advice to prescribers, not evidence of assessment. The labels do assert "kidney toxicity was not observed in the clinical studies", which is a statement about findings, but no endpoint table stands behind it. Now asserted_negative_regulatory — stronger than silence, weaker than a measured result. MSR016 (inclisiran) reclassified on the same basis.

Implemented as scripts/add_negative_eligibility.py, adding two derived columns and touching no source assertion:

- negative_eligibility — positive_finding | confirmed_negative | asserted_negative_regulatory | efficacy_derived_negative | not_eligible_negative
- nephrotox_grade_modeling — equals nephrotox_grade **except blank** where the negative is not confirmed. Missing, not zero. **This is the column to train on.**

nephrotox_grade is preserved unchanged on all 246 rows, verified by the release gate.

Result: of 21 clinical grade-0 rows, exactly one is a confirmed negative — MSR066 (cemdisiran, quantitative eGFR -2.9 vs placebo -6.3 mL/min/1.73 m², primary source read). 20 rows are gated out of negative training.

Priority 3 — comparability and the bridge → **ACCEPTED; found a real bug**

Normalisation. Already implemented before this review: readout_unit distinguishes pct_saline_control from pct_compound_1-1_reference, and both extractors write a do-not-pool note per row. I consider this closed and did not change it.

Bridge confounding. Beebop is right that a max-grade comparison across different studies confounds species with dose, assay, follow-up and ascertainment. Added comparison_type: **10 of 15 comparisons are paired_same_source** (human and animal evidence from the same document); 5 are cross_study_overlap and are hypothesis-generating at best.

Unsupported over-prediction claims. Beebop's closure check — "unsupported human negatives cannot produce claims that animals overpredict toxicity" — had real teeth. **Five of my six animal_over_predicts verdicts rested on human negatives that fail eligibility.** Those are now indeterminate_human_negative_unsupported; the verdict is suppressed rather than published. One over-prediction verdict remains. The two *under*-predictions (inotersen, givosiran) are unaffected because they rest on positive human findings, not negatives.

Bug found while implementing this. My "same source" test initially failed on the N3 patent because the data cited it two ways — US11105794 for Table 2 and US11105794B2 for Table 1 — so the patent did not match itself and its three compounds were wrongly classed cross-study. Normalised 21 rows to one spelling (source_table still distinguishes the tables), which moved paired comparisons from 7 to 10.

Statistics. **Beebop is right and I was wrong.** I had claimed the provenance/outcome confound "weakened 3.7×", which is a **p-value ratio** — not an effect size, and it moved largely because I had added three rows to the anchored arm's denominator. Recomputed on the honest measure, the risk difference moved **57.9 → 50.0 percentage points**: a real but modest 7.9-point reduction, not a 3.7-fold one. Corrected in all five documents that carried the claim (NARRATIVE, METHODOLOGY_PHASE2, STATUS, schema, CLINICAL_VALIDATION).

Priority 4 — characterization and release documents → **ACCEPTED**

Purity wording. Accepted. Replaced "cannot be curated" / "cannot be closed by further curation" with the narrower supported statement: not reported in the sources reviewed, with the search that was actually run named (both patents searched for purity / HPLC / UPLC / LC-MS / mass-spec; labels and trial papers do not publish per-batch purity), and an explicit note that targeted supplements and manufacturer batch records **remain unexamined**. I have *not* contacted any author or manufacturer; that is listed in §6 for Oscar's decision.

Document reconciliation. Done — stale merged-view dimensions, bridge figures and sequence counts corrected; new fields and the register documented in `schema.md`.

Release identifier. Accepted from "Common gaps". Added `scripts/make_release_manifest.py` → `RELEASE_MANIFEST.json`, binding the canonical tables, derived files and documents to `kidney-5d82de234c71` with SHA-256 per file. The id is content-derived, so identical data always yields the same id and any table change changes it — a reviewer holding an older Drive export can tell immediately.

Workbook order. Implemented Beebop's suggested order, with the animal evidence moved to a genuinely separate appendix rather than merely labelled:

Summary → 1 Human trials (verified) → 2 Human trial measurements → 3 Human lab evidence → 4 German's analysis → 5 Oligos and characterization → 6 Data dictionary → A1 APPENDIX animal evidence → A2 APPENDIX all measurements → A3 APPENDIX human vs animal

The release gate now asserts that every human/design tab precedes every appendix tab.

Nothing rejected

I rejected none of the four priorities. The one I pushed back on in part is Priority 2's framing: "recheck the remaining 29" is better scoped as "recheck the 8 that assert a negative", since 21 of the 29 are positive findings where the ascertainment question does not arise.

3. Implemented changes

File	Status	Effect
<code>scripts/build_study_register.py</code>	new	Per-row study attribution + 7 counting rules → trial counts
<code>data/clinical_study_register.csv</code>	new	42 clinical rows resolved to 35 distinct studies
<code>scripts/add_negative_eligibility.py</code>	new	<code>negative_eligibility</code> + <code>nephrotox_grade_modeling</code>
<code>scripts/make_release_manifest.py</code>	new	Content-derived release id + checksums
<code>RELEASE_MANIFEST.json</code>	new	Binds data, derived files and documents to the release
<code>scripts/split_human_animal.py</code>	modified	<code>comparison_type</code> , verdict suppression, <code>human_negative_eligible</code>
<code>scripts/build_merged.py</code>	modified	Carries the two new columns (46 cols)
<code>scripts/build_workbook.py</code>	modified	Human-first tab order; animal appendix; new columns
<code>scripts/release_check.py</code>	modified	Gates trial counting, eligibility coupling, tab order
<code>data/measurements.csv</code>	modified	+2 derived columns; 21 <code>source_ref</code> normalised; notes appended
<code>NARRATIVE.md</code>	modified	Human-trials-first presentation; bridge and statistics corrected
<code>METHODOLOGY.md</code> · <code>METHODOLOGY_PHASE2.md</code> · <code>STATUS.md</code> · <code>schema.md</code> · <code>CLINICAL_VALIDATION.md</code> · <code>PADP.md</code>	modified	Statistics, purity wording, bridge figures, new fields

Historical records preserved. No measurement row was deleted, no grade overwritten, no animal data removed. Every reclassification is an added derived column with its reason recorded in that row's notes. `BEEBOP_SUGGESTIONS_2026-09-30.md` was not edited.

4. Counts, by evidence class

Counting rule for trials: a trial counts **once** however many measurement rows, publications, labels or repeated outcomes reference it. An extension or longer follow-up of the same trial is the same trial. A regulatory label, review, case report or pooled cross-trial analysis is **not** a trial. A study with neither a name nor a registry identifier is unresolved and does not count. No identifier was invented.

Human clinical evidence (deduplicated by study)	count
Verified trials (identified and primary document read)	3
Trials identified but not yet read	14
→ total distinct trials identified	17
Distinct regulatory labels (not trials)	13
Pooled cross-trial analyses	1
Reviews / editorials	2
Case reports	1
Unresolved citations	1
Clinical measurement rows (<i>not a trial count</i>)	42
Unique compounds with clinical evidence	31

Verified trials: **DMD114673, Cemdisiran phase 2 IgAN, SEQUOIA.**

Other evidence classes	count
Human laboratory / ex-vivo rows (separate; not trials)	67
Animal appendix — animal in-vitro	81
Animal appendix — animal in-vivo	56
→ animal total, excluded from all human totals	137
All measurements	246
Oligos / with sequence	65 / 55
Rows gated out of negative training	20
Clinical grade-0 rows that are confirmed negatives	1

5. Additional findings

- 1 **The source_ref inconsistency** (§2, Priority 3) — the same patent cited two ways, silently breaking any same-source analysis. Found only because the pairing test produced a result I did not believe.
- 2 **My own internal inconsistency on MSR045** — documented as efficacy-derived in one file, classified as a measured negative in another. Worth noting as a class of error: a finding recorded in prose does not propagate itself into the data.
- 3 **Priority 2's scope is 8 rows, not 29** — 21 of the 29 are positive findings.
- 4 **The release gate caught my own tab rename** before I shipped it, which is the argument for having it.
- 5 **Statistical framing matters more than I treated it.** The "3.7×" claim was not a small wording issue; it misrepresented a 7.9-point effect as a 3.7-fold one.

6. Decisions requiring German / Oscar

For German (scientific):

- 1 **Are the five asserted_negative_regulatory downgrades right?** Does a label's "kidney toxicity was not observed in the clinical studies" — with no endpoint table — support a usable safety negative, or not? This currently gates 6 rows out of modelling.
- 2 **Is MSR066 genuinely the only confirmed clinical negative?** It is the only one with a quantitative safety endpoint from a document we have read. If that is right, the dataset has essentially no usable clinical negative class, which is a significant statement.
- 3 **Does same-patent-different-table count as genuinely paired?** The three Roche compounds have human cell data in Table 2 and rat in-vivo data in Table 1 of one patent. I class them paired; they are the same laboratory and compounds but different experiments.
- 4 **The two grading rubrics** (NARRATIVE §4.3) — thresholds are anchored on each source's own innocuous control, but the cut-points are mine.
- 5 **Whether 3 verified trials is the right headline**, or whether identified-but-unread trials should be reported as a second headline tier.

For Oscar (access / permissions):

- 1 **Author or manufacturer contact for purity.** Per-batch purity exists only in Certificates of Analysis. I have not contacted anyone and will not without your approval.
- 2 **The 14 identified-but-unread trials.** Eleven are open-access PMC articles that NCBI blocks automated retrieval of but that open normally in a browser; the rest are paywalled or FDA review documents. Listed in SOURCES_TO_ACQUIRE.md.
- 3 MSR064 **zilebesiran** cites KARDIA_trials, which is not a resolvable citation. It needs a real reference before anyone can verify that row.

7. Verification, limitations, readiness

Verification performed. scripts/release_check.py passes, now gating: integrity (65 oligos / 246 measurements, 0 orphans, no duplicate keys, grades in range); strict-kidney isolation (246/246 is_kidney_specific=TRUE, renal tissues only, no hepatic readouts, models or hepatotoxicity source panels, viability rows renal-only); the register covering every clinical row with no non-trial class counting toward the trial total; a deduplicated trial count strictly below the row count; nephrotox_grade_modeling blank **iff** the negative is not eligible (0 mismatches); nephrotox_grade preserved on every row; all human/design tabs preceding the animal appendix; and every Human-trials row carrying both a sequence cell and a grade. All derived files regenerated through the existing scripts.

Limitations.

- Only **3 of 17** trials have had the primary document read. The headline verified count is genuinely small and should not be inflated.
- **One** confirmed clinical negative. The clinical negative class is effectively unusable for modelling as it stands.
- Purity unreported for all 65 oligos; targeted supplements and batch records unexamined.
- All grades remain **provisional** pending German's sign-off.
- 39 of 246 rows lack a numeric dose, pending FDA Pharmacology/Toxicology reviews.
- The bridge is 15 compounds, 10 genuinely paired — descriptive, not a fitted species model.
- Scope confirmed kidney-only; no other endpoint's data was read or written.

Readiness. The dataset and its four submission artefacts are **structurally complete and internally consistent at release** kidney-5d82de234c71, and the human-first presentation Oscar asked for is implemented rather than described. It is **not scientifically signed off**: the trial verification tier, the negative-eligibility downgrades and the

grading thresholds are curation judgements awaiting German. The honest summary is that this round made the dataset's weaknesses legible and machine-readable rather than making them smaller — the trial count fell from an implied 42 to a verified 3, and the confirmed clinical negatives from 21 to 1. Both are corrections, not regressions.